DIGITAL LOGIC DESIGN • FINAL PROJECT

# Design & Simulation of an Automatic Street Light Control System

Lamp ON in the dark, OFF in daylight, simulated in Logisim

# Introduction and Objective



## Introduction

- Conventional street lights can stay ON even in daylight, wasting electricity.
- An automatic system uses a light sensor (LDR) to sense brightness.
- A comparator decides if it is dark and switches the lamp ON or OFF.
- Logisim is used to simulate this logic before building real hardware.
- Handor timerswitchingcannotfollowcloudydays or early evenings.

## Objectives

- Switch the lamp automatically based on light level.
- Reduce unnecessary ON-time and energy use.
- Design the control logic with digital components.
- Verify correct behaviour through simulation.
- Remove theneedformanual switching or timers.
- Keep the design simple: one comparator and one constant.
- Make the dark/bright threshold easy to change.
- Show the lamp state clearly with an LED output.

**Core rule:** Light level < Threshold → DARK → Lamp ON | Light level ≥ Threshold → BRIGHT → Lamp OFF

# Theory and Logic Circuit






## Concept and logic

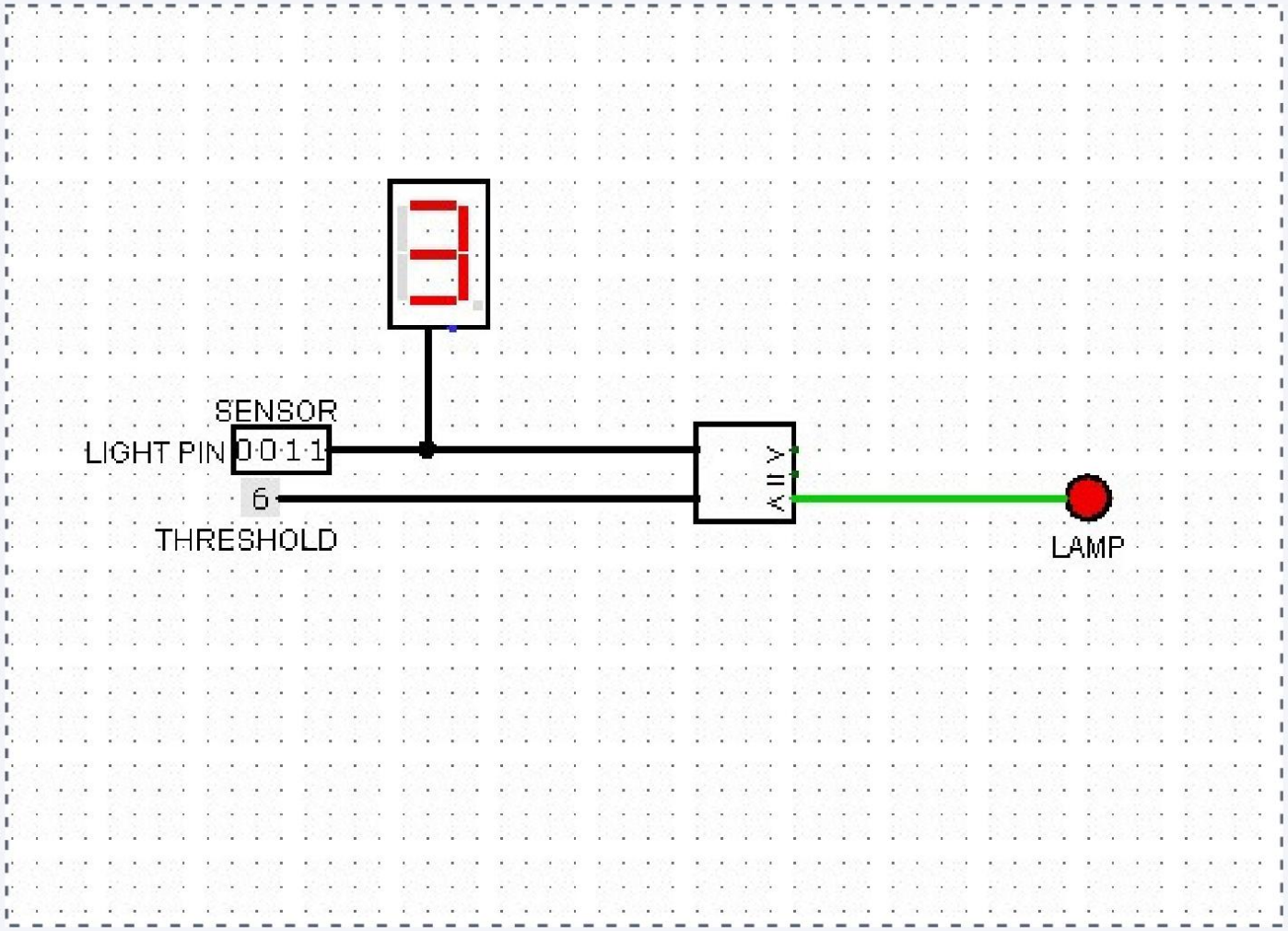
- LDR resistance drops in light and rises in dark.
- Simulated as a 4-bit number: 0 = darkest, 15 = brightest.
- Comparator output A < B is 1 when LIGHT < 6.
- Logic: LAMP = (LIGHT < THRESHOLD)
- 1-bit logic: A<B = A′·B (NOT gate + AND gate).
- 6= 0110: levels0 to5aredark,6 to15 arebright.
- Bits are compared from the MSB down; first difference decides.
- Equal to the threshold counts as bright (lamp OFF at 6).

### 1-bit comparator (A < B)

| A | B | A < B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

# Circuit Design in Logisim






## Components used

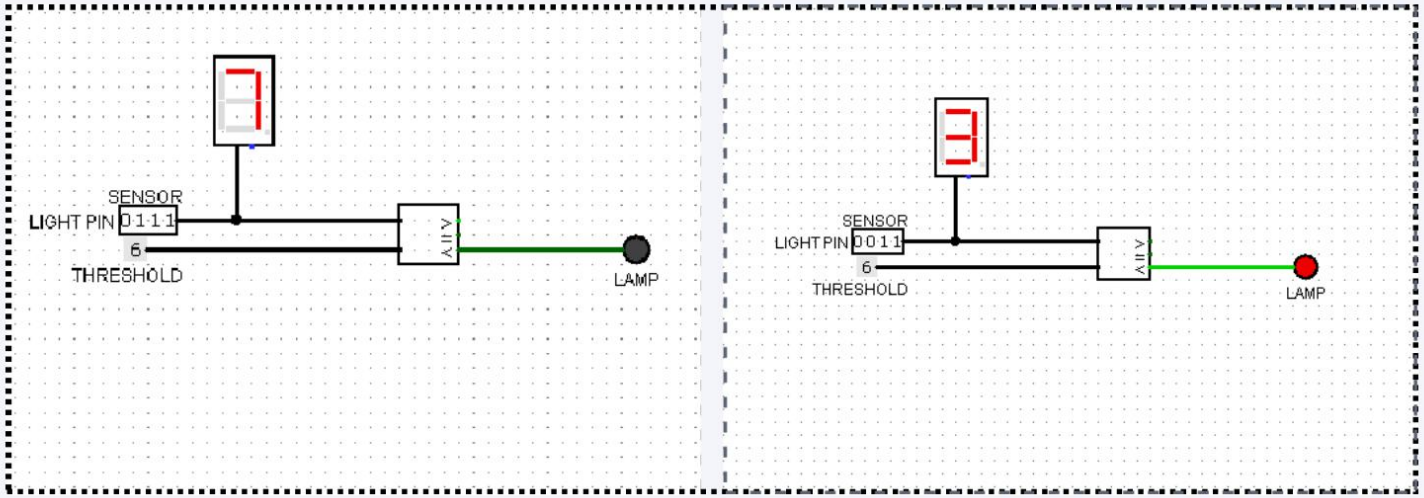
- Pin (4-bit) labelled LIGHT
- Constant 6 (4-bit)
- Comparator (4-bit, Unsigned)
- LED labelled LAMP
- Hexdisplay showingthe LIGHT value
- Wires: 4-bit buses, 1-bit output

## Working

- LIGHT and 6 enter the comparator.
- The “<” output goes to the LED.
- LIGHT < 6: output 1, lamp ON.
- LIGHT ≥ 6: output 0, lamp OFF.
- Click the pinto change LIGHT.
- LED is red when ON, dark when OFF.
- Unsigned keeps 8 to 15 positive.

# Simulation Results






*Note: comparator Numeric Type is set to Unsigned so 12 is read as 12, not −4.*

**Observation**
The lamp turns ON for every input below 6 and OFF for 6 and above, matching the expected behaviour.
The lamp state flips exactly between LIGHT=5 and LIGHT=6.

**Truth table (threshold = 6)**

| LIGHT | Binary | A<B | LAMP |
|---|---|---|---|
| 0 | 0000 | 1 | ON |
| 2 | 0010 | 1 | ON |
| 5 | 0101 | 1 | ON |
| 6 | 0110 | 0 | OFF |
| 172 | 1100 | 0 | OFF |
| 15 | 1111 | 0 | OFF |

# Conclusion and Applications



## Outcome

- Built a working automatic street light controller in Logisim.
- Lamp switches correctly for dark and bright inputs.
- Threshold logic verified with a truth table.
- Tested with a darkinput(2) and a bright input (12).
- Uses only one comparator, one constant and one LED.
- Same rule holds for all 16 light levels.
- Shows how basic digital blocks solve a real problem.

## Applications

- Street and highway lighting.
- Garden, campus and parking lights.
- Smart-city energy saving.
- Home security lights.
- Busstops andrailway platforms.
- Industrial yards and warehouses.
- Signboards and hoardings.
- Tunnel and bridge entrances.
- Rural roads and village lighting.

## Improvements

- Add a motion (PIR) sensor for full brightness.
- Add hysteresis to stop flicker at dusk.
- Use solar power with battery.
- Build a hardware prototype with LDR and Arduino.
- Make the threshold adjustable with a second input pin.
- Average several LDRs for a steadier reading.
- Add a manual override switch.
- Add a lamp-fault alarm.

# Team Members



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